

README - R/utils/

C.J. Tinant

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1 Utilities (R/utils/)

This folder contains helper scripts and functions used across the project for spatial data processing, zonal summaries, and other repetitive tasks. Functions here are **modular** and **reusable**, but are not meant to be run as standalone workflows.

The goal is to keep scripts DRY (“Don’t Repeat Yourself”) and consistent across different processing steps.

1.1 Usage

Functions in these scripts should be sourced in workflow scripts:

```
source(here::here("R/utils/metadata/extract_full_metadata.R"))
```

- Spatial helpers are intended to be written to be format-agnostic where possible (e.g., can accept both sf and terra objects).
- Spatial helpers are intended to return objects ready for use with {terra} workflows.

1.2 Folder Organization

- **metadata/** — Functions for extracting, summarizing, and cleaning metadata from raster and vector files.
- **inventory/** — QA utilities for file inventory, duplicate detection, and archive management.
- **spatial/** — Spatial preprocessing tools (e.g., reprojection, geometry cleanup, resource downloads).
- **qa/** — Code quality and style check scripts.

```

R/
utils/
  inventory/          # Inventory scripts for QA and archiving
    inventory_feature_data.R
    inventory_raster_data.R
    verify_prism_archive.R
    move_zip_to_archives.R

  metadata/          # Functions for metadata extraction and cleaning
    extract_full_metadata.R
    extract_summary_metadata_raster.R
    extract_summary_metadata_vector.R
    parse_metadata.R
    metadata_manual_cleaning_tools.R
    summarise_raster_crs.R
  qa/                # Code quality utilities
    run_linr_on_a_directory.R

  spatial/           # Spatial processing utilities
    assert_inputs_ok.R          # Pre-flight checks for zonal aggregation
    continuous_rast_summ_by_zone.R # continuous raster summaries by zone
    download_ecoregion_resources.R # DL EPA ecoregion layers, metadata
    prep_and_align.R             # Align polygon zones to raster CRS
                                   # and prepare a raster for zonal work
    process_geometries.R          # Adds columns for centroid coordinates
    reproject_bil_to_epsg5070.R
    vector_helpers.R

README.Rmd

```

1.2.1 Unit Testing

(future) For consistent unit testing, see the `tests/` folder and the `testthat` examples in the project root.

1.3 Naming conventions

- snake_case for function names and arguments.
- Keep function scope narrow — one main purpose per function.
- Avoid hard-coding file paths or project-specific constants; pass them as arguments.

1.4 Maintenance

- If a helper is updated, check all scripts that use it to avoid breaking changes.
 - Add a brief header comment to each helper file:
 - Script Name
 - Author
 - Last Updated
 - Purpose
 - Dependencies
-

1.5 Notes

- Keep each function focused on a single responsibility.
 - Avoid hard-coded paths; use `here::here()` for project portability.
 - Update this README when adding, renaming, or relocating utility scripts.
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